8	SAY WHAT YOU DO	6	5	4	3	DO WHAT YOU SAY	1

OPERATING CONDITIONS

NORMAL	11,714 HP [8,735 kW] @ 9,340 RPM
NORMAL TORQUE	79,044 IN-LB [8,930 N-m]
MAX POWER	14,057 HP [10,482 kW] @ 9,807 RPM
MAX POWER TORQUE	90,338 IN-LB [10,206 N-m]
TRIP SPEED	10,788 RPM
SELECTION TORQUE (INCLUDES 1.50 S.F. ON NORMAL TORQUE)	118,566 IN-LB [13,396 N-m]

COUPLING RATING

MAX CONTINUOUS TORQUE	120,000 IN-LB [13,558 N-m]
PEAK TORQUE	158,000 IN-LB [17,851 N-m]
MAX MOMENTARY TORQUE	210,000 IN-LB [23,726 N-m]
API 671 RATED SPEED	12,000 RPM

RIGID/SHAFT INFORMATION	DRIVING	DRIVEN
TORQUE CAPACITY (BASED ON MU = 0.15)	248,500 IN-LB [28,076 N-m]	247,600 IN-LB [27,974 N-m]
REQUIRED MOUNT AND DISMOUNT PRESSURE	26,200 PSI [180,643 kPa]	26,200 PSI [180,643 kPa]
MAX ALLOWABLE ASSEMBLY PRESSURE	32,600 PSI [224,769 kPa]	32,500 PSI [224,080 kPa]

MASS ELASTIC DATA

COMPONENT WEIGHT

	DRIVING	DRIVEN
RIGID (BORED)	23 LB [10.4 kg]	23 LB [10.4 kg]
ADAPTER (INCL. 1/2 DISC PACK)	9 LB [4.1 kg]	9 LB [4.1 kg]
SLEEVE (INCL. 1/2 DISC PACK)	9 LB [4.1 kg]	9 LB [4.1 kg]
SPACER	7 LB [3.2 kg]	
TOTAL COUPLING WEIGHT (BORED)	89 LB [40.4 kg]	
MOMENT OF INERTIA (WR^2)	800 LB-IN^2 [0.23kg-m^2]	
TORSIONAL STIFFNESS (BASED ON 1/3 SHAFT PENETRATION)	4.0 X 10^6 IN-LB/RAD [0.45 X 10^6 N-m/RAD]	
DISC PACK ANGULAR STIFFNESS	1,800 IN-LB/DEG [203 N-m/DEG]	
LATERAL CRITICAL SPEED (SIMPLY SUPPORTED TUBE)	110,800 RPM	

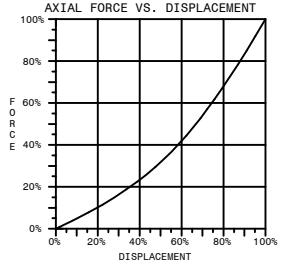
	ANGULAR	AXIAL
MISALIGNMENT: MAX CONTINUOUS	0.25 DEG	±0.120 [3.05]

NOTE: TRANSIENT AXIAL MISALIGNMENT CAN BE 133% OF MAX CONTINUOUS

DISC PACK ℄ TO ℄ DISTANCE 16.5 [419]

EQUIVALENT PARALLEL OFFSET .072 [1.83]

AXIAL FORCE VS. DISPLACEMENT



AXIAL DISPLACEMENT ±(IN)	±[mm]	(% OF CONT.)	FORCE (LB)	[N]
.120	3.05	100	644	2,865
.060	1.52	50	183	814
.012	0.30	10	28	125
.000	0.00	0	0	0

PRODUCTION DRAWINGS DRAWING NO.	DESCRIPTION
CC69883	SHIM PACK
CC68493	DISC PACK
1157867	ASSEMBLY DWG
ES-030-BL	BALANCE SPEC

LIST OF MATERIAL AND DRAWINGS FOR ONE COUPLING

QTY	PART NO.	PART DESCRIPTION	MATERIAL	MK NO.
1	1164653	RIGID, DRIVING	AISI 4340 HT	1
1	1165483	RIGID, DRIVEN	AISI 4340 HT	2
2	CC70170	ADAPTER	AISI 4340 HT	3
2	1167490	SLEEVE(L=4.957[125.91])	AISI 4340 HT	4
1	1167494	SPACER(L=5.906[150.0])	AISI 4140 HT	5
2	CX99928	DISC PACK ASSEMBLY	301 STAINLESS STL	6
16	WA82275	7/16-20 NF x 2.00 CAPS	ALLOY STEEL	7
16	WA82243	7/16-20 NF LOCKNUT	STEEL	8
16	WA80623	7/16 HARDENED WASHER	ALLOY STEEL	9
□ 44	WA82271	5/16-24 NF x 1.30 CAPS	ALLOY STEEL	10
□ 44	WA82241	5/16-24 NF LOCKNUT	STEEL	11
2	FX10140	.060 SHIM PACK	STEEL	12
* 8	WA81957	1/4-28 NF x 1.50 SHCS	ALLOY STEEL	13
* 8	WA82011	3/8-24 NF x 1.50 S.S.	ALLOY STEEL	14

□ - QUANTITY INCLUDES AN ADDITIONAL 10%
* - INDICATES 'SHIPPING SCREWS' NOT SHOWN.
'SHIPPING SCREWS' - YELLOW COLLAPSING CAPSCREWS
RED LOCKING SETSCREWS
NOTE: ALL 'SHIPPING SCREWS' MUST BE REMOVED AT FINAL ASSEMBLY.
(SEE INSTALLATION INSTRUCTION 16-602)

APPROVAL REQUIRED

THIS DRAWING IS NOT APPROVED FOR MANUFACTURING

1. APPROVED. □
2. APPROVED AS NOTED. □
3. REVISE AS NOTED AND RESUBMIT. □
4. REJECTED. □

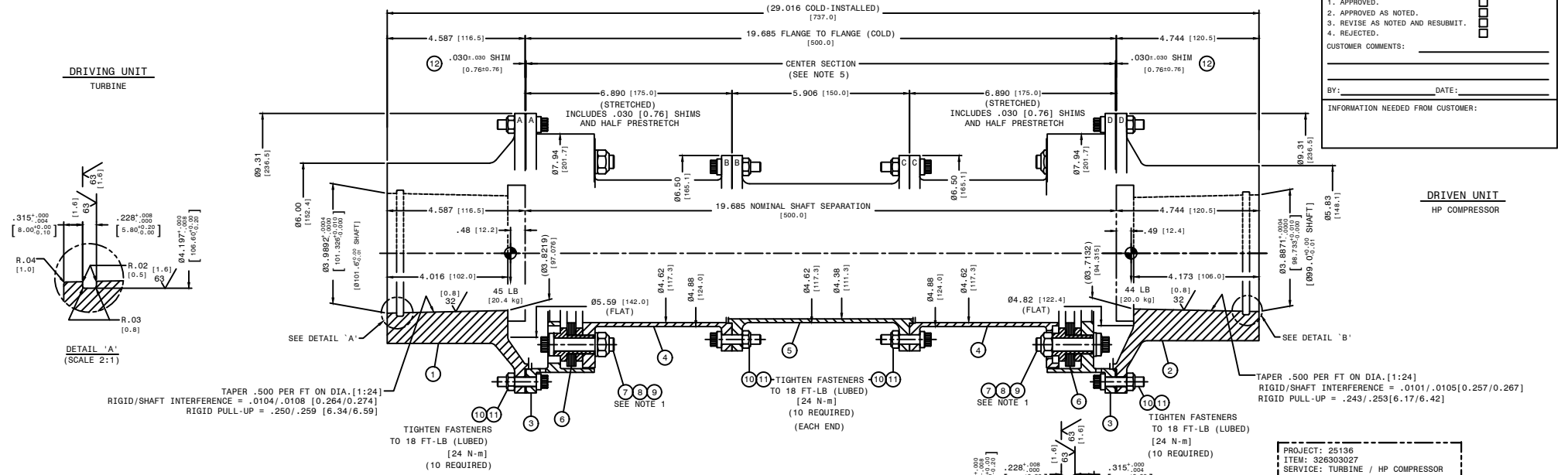
CUSTOMER COMMENTS:

BY: DATE:

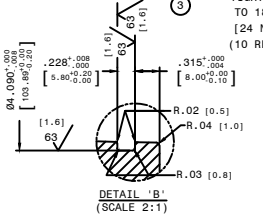
INFORMATION NEEDED FROM CUSTOMER:

DRIVING UNIT
TURBINE

DRIVEN UNIT
HP COMPRESSOR



DETAIL 'A'
(SCALE 2:1)



DETAIL 'B'
(SCALE 2:1)

NOTES:

1. KOP-FLEX DISC COUPLINGS ARE PROVIDED AS FACTORY ASSEMBLED UNITS. WARNING: DO NOT TIGHTEN OR LOOSEN DISC PACK BOLTS AND NUTS.
2. COUPLING IS DESIGNED IN ACCORDANCE WITH API 671, WITH STANDARD EXCEPTIONS PER KOP-FLEX FORM #16-1009.
3. COUPLING TO BE DYNAMICALLY BALANCED PER API 671. (COMPONENT BALANCE AND ASSEMBLY CHECK)
4. ✥ INDICATES EFFECTIVE CENTER OF GRAVITY OF HALF COUPLING.
5. CENTER SECTION IS DESIGNED TO STRETCH FLEXIBLE ELEMENTS AT INSTALLATION. NOMINAL CENTER SECTION FREE LENGTH (UNSTRETCHED) = 19.578 [497.28] SHIM AS REQUIRED TO OBTAIN PRESTRETCH VALUE OF .047 [1.20] AXIAL GROWTH (.047 [1.20]) - PRESTRETCH (.047 [1.20]) = RUNNING NEUTRAL
6. DRIVING AND DRIVEN RIGID BORES PER NEW JCM 719-10360 Rev. 7.

BALANCE REQUIREMENTS BALANCE CRITERIA (METHOD 2)	RIGID (DRIVING END)	RIGID (DRIVEN END)	SPACER	CENTER ASS'Y (DVG END)	CENTER ASS'Y (DVN END)	DOES NOT CONTAINS DISC PACK WEIGHT ADAPTER (DVG & DVN)	SLEEVE (DVG & DVN)
COMPONENT BALANCE	.52 g-in [13.2 g-mm]	.52 g-in [13.2 g-mm]	.28 g-in [7.2 g-mm]	.50 g-in [12.7 g-mm]	.48 g-in [12.1 g-mm]	.28 g-in [7.2 g-mm]	.28 g-in [7.2 g-mm]
	DRIVING END	DRIVEN END					
ASS'Y CHECK BALANCE	10.21 g-in [259.23 g-mm]	9.98 g-in [253.47 g-mm]					

PROJECT: 25136
ITEM: 326303027
SERVICE: TURBINE / HP COMPRESSOR

CUSTOMER: NEW JCM
INQUIRY NO.- BAL NO. 11218172A
CUSTOMER'S ORDER NO.- 2025-12-P25136
KOP-FLEX'S CP- 28428408
QUANTITY- (2) COUPLINGS + (2) FLANGE ACCESSORY SET (MK#10-11)

ZONE	REV	REVISION DESCRIPTION	BY	DATE	CHK	APPR
	A	INITIAL RELEASE FOR APPROVAL	JNW	11/22/21	VKG	VKG

CAP JD YES

DWN - DATE - CHK - DATE - ENG - DATE - PROJECTION - SCALE TO SCALE APPR - DATE -

DO NOT SCALE DRAWING. UNLESS OTHERWISE SPECIFIED: ALL DIMENSIONS IN INCHES, DIMENSIONS IN BRACKETS [ ] ARE IN MM. BORE AND KEYWAY TOLERANCE PER ABMA 9002 OR ABMA 9112.

Regal Beloit America, Inc.

Quality Management System Certified to ISO 9001

KOP-FLEX

TITLE: KOP-FLEX™ #254 MP DISC COUPLING

DWG. NO. 1174836 SHEET 1 OF 1 REV A

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